

Ashok Gahatraj

Geographer | Geospatial Data Analyst

ashoker@nmsu.edu | Las Cruces, NM | [Linkedin](#) / [GitHub](#) / [Website](#)

WORK EXPERIENCE

Graduate Researcher | New Mexico State University

Aug. 2024 – Present

Las Cruces, NM

- Built a MODIS/Landsat phenology pipeline (R + GEE), improving multi-year vegetation dynamics extraction across three major North American deserts, revealing key climate-biome interactions.
- Processed time-series datasets (>200 GB) and automated workflows using Python and R to improve efficiency.

GIS Analyst | New Mexico State University

May. 2025 – Aug. 2025

Las Cruces, NM

- Led the design and deployment of a campus-wide ArcGIS Indoors Information Model, architecting the data schema and integrating complex floor plans to enhance research space management and utilization for university stakeholders.
- As a GIS Analyst, I have.
 - Built dynamic dashboards for occupancy, funding, and space utilization used by university leadership.
 - Built a campus Indoors Hub site combining indoor maps, wayfinding tools, and analytics for digital twin visualization.

Data Analyst | Numeric Mind

Oct. 2022 – Jun. 2024

Kathmandu, Nepal

- Automated data cleaning and validation processes for client datasets using Python and SQL, improving efficiency.
- Collaborated with project managers and engineers to create spatial and analytical tools for business intelligence needs.

EDUCATION

New Mexico State University | Expected May 2026

Master's in Applied Geography | Las Cruces, NM

- **GPA: 3.97/4.0**

CERTIFICATIONS, SKILLS & INTERESTS

- **Certifications:**
 - FAA Part 107 Certified Drone Pilot
 - Google Data Analytics Specialization (Coursera)
 - Introduction to GIS Programming – UT Knoxville
- **Technical Skills:**
 - GIS & Remote Sensing: ArcGIS Pro, ArcGIS Indoors, QGIS, ENVI, Agisoft Metashape, Google Earth Engine
 - Programming & Analytics: Python, R, SQL
 - Surveying & UAV: Total Station, DGPS, UAV Mapping & Photogrammetry
- **Interests:** Hiking; Soccer; Traveling; Photography; Formula One